

Nosherwan Babar

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Professional Summary

Software engineer who builds production AI services and delivers them directly with customers. Shipped 25+ Python and JavaScript services across 4 client engagements in an embedded, consulting-style role, then deployed and monitored ML across 7 production systems on GCP at URBN. Built and still operates CyberSentinel, a multi-agent LLM platform with RAG, a 332-test evaluation suite, and responsible AI safeguards as release gates. MS Data Science, Drexel University, September 2026.

Core Competencies

Production Software Engineering | Customer Co-Engineering | Responsible AI & Safeguards | LLM Solutions (RAG, Prompt Engineering, Model Tuning) | Evaluation & Observability | Python & JavaScript | Automated Testing

Work Experience

Machine Learning Engineer Co-op URBN (Urban Outfitters, Inc.)

September 2025 – March 2026
Philadelphia, PA

- Deployed and monitored ML services across 7 production systems on GCP, serving real-time inference through FastAPI on Cloud Run with observability, drift detection, and performance monitoring.
- Fine-tuned transformer models in PyTorch on Vertex AI and built the embedding generation and inference paths behind them.
- Built Airflow pipelines with schema-validation gates for data quality, running ingestion, evaluation, deployment, and drift-triggered retraining, and cut manual intervention by roughly 70%.
- Wrote operational runbooks for pipeline failures and maintained CI/CD with automated tests to improve reliability.
- Worked with data scientists and product stakeholders to turn requirements into technical designs, and explained model tradeoffs and success metrics to non-technical audiences.

Associate Software Engineer Buggcy

January 2023 – August 2024
Lahore, Pakistan

- Shipped 25+ Python and JavaScript backend services and REST APIs across 4 client engagements, embedded end-to-end with each customer team from requirements through production.
- Contributed to architecture and code reviews, and wrote API documentation thorough enough for customer teams to onboard from specs alone.
- Cut API latency 15% through SQL query optimization, with enforced testing on every service.

Projects

CyberSentinel – Multi-Agent LLM Platform (deployed) | *cybersentinel-prod.web.app*

Production multi-agent platform that answers security questions via retrieval-augmented generation over live CVE/NVD data, deployed on GCP Cloud Run.

- Built a 5-agent LangGraph and LangChain platform with routing and reusable components shared across agents (prompt engineering templates and tool definitions), containerized with Docker.
- Built retrieval-augmented generation over live data: chunking, embeddings, and vector retrieval from ChromaDB, with context assembly tuned for structured JSON output.
- Evaluated model risks with a 332-test evaluation suite that scores every output on relevance, accuracy, and safety as a CI/CD release gate, plus tracing for latency and cost as ongoing safeguards.

Parameter-Efficient Fine-Tuning Research | *MS Thesis (DSCI 690), Drexel University, 2026*

Benchmark study comparing model tuning methods against full fine-tuning on domain question answering.

- Compared LoRA, adapter, and prefix tuning against full fine-tuning of Flan-T5-base and Llama-3.2-3B on medical and legal QA benchmarks, building reproducible evaluation that scored accuracy against parameter and compute cost.

Education

Master of Science in Data Science – Drexel University

Philadelphia, PA | GPA: 3.75/4.00 | Applied ML, NLP with Deep Learning, Recommender Systems, Quantitative Data Analysis

September 2026

Bachelor of Science in Computer Science – University of Central Punjab

Lahore, Pakistan | GPA: 3.41/4.00

July 2023

Skills

Languages: Python, JavaScript, SQL, Shell

AI & LLM: LangGraph, LangChain, RAG, Embeddings, Vector Databases (ChromaDB), Prompt Engineering, Model Tuning (PEFT/LoRA), Evaluation Frameworks, Guardrails, LLM APIs (Gemini, OpenAI)

Engineering: FastAPI, REST APIs, Docker, Git, GitHub Actions (CI/CD), pytest, Code Reviews, Airflow

Cloud & ML: GCP (Vertex AI, Cloud Run, BigQuery), PyTorch, scikit-learn, Observability, Drift Detection